History of AI

Artificial Intelligence did not develop in one day. It developed gradually over many years through the work of mathematicians, scientists, and computer researchers.

1940s – Beginning of the Idea

Scientists started thinking about whether a **machine could behave like a human brain**.

In **1943**, Warren McCulloch and Walter Pitts developed a mathematical model of an **artificial neuron**. This became an important foundation for neural networks.

example:
A human brain has neurons that process information. Researchers wanted computers to work in a somewhat similar way.

1950 – Alan Turing

In **1950**, British mathematician **Alan Turing** asked an important question:

"Can machines think?"

He proposed the **Turing Test**.

The basic idea is that if a person communicates with a computer and cannot reliably determine whether they are communicating with a human or a machine, the machine may be considered intelligent.

Example:
Imagine chatting with two people through text. One is a human and one is a computer. If you cannot tell which is the computer, the computer has demonstrated a form of machine intelligence.

1956 – Birth of AI

The year **1956** is considered a major milestone in AI history.

At the **Dartmouth Conference**, John McCarthy and other researchers formally promoted the study of **Artificial Intelligence**.

John McCarthy is widely known for **coining the term "Artificial Intelligence."**

The main goal was:

To make machines perform tasks that require human intelligence.

1960s – Early AI Programs

During the 1960s, researchers developed programs that could perform tasks such as:

a. Solving mathematical problems

b. Playing games

c. Proving simple mathematical theorems

d. Understanding limited natural language

At this stage, AI systems mainly followed **rules programmed by humans**.

1970s–1980s – Expert Systems

Researchers began developing **Expert Systems**.

An expert system stores knowledge and rules from human experts and uses them to make decisions.

Example:
A medical expert system could use rules such as:

IF a patient has fever and certain symptoms
THEN suggest possible diseases.

These systems were useful in areas such as medicine, business, and engineering.

1980s–1990s – Machine Learning

AI gradually moved from systems that only followed fixed rules toward **Machine Learning**.

Machine learning allows a computer to **learn patterns from data**.

Example:

Instead of programming:

"If an email contains these exact words, call it spam."

We can give the computer thousands of examples of **spam and non-spam emails**, and it learns patterns to identify new spam emails.

2000s – Big Data and Better Computers

AI developed rapidly because of:

- Large amounts of digital data
- Faster computers
- Better algorithms
- Improved storage
- Internet growth

More data allowed computers to learn more complex patterns.

2010s – Deep Learning

Deep Learning, a branch of machine learning based on multi-layer neural networks, became extremely successful.

AI systems became much better at:

- Image recognition
- Speech recognition
- Language translation
- Playing games
- Understanding text

For example, computers became capable of recognizing objects in photographs and understanding spoken commands.

2020s – Generative AI

In the 2020s, **Generative AI** became widely used.

Generative AI can create new content such as:

- Text
- Images
- Audio
- Video
- Computer programs